IVB.Tech I Semester (E.C.E)

23A04701	**DATA COMMUNICATIONS AND NETWORKING**	**L**	**T**	**P**	**C**
		3	0	0	3

Course Objectives:

1. To provide a conceptual understanding of the fundamentals of data communications and computer networks.
2. To explore different network architectures, models, and transmission media used in data communication.
3. To analyze error detection and correction methods, data link protocols, and medium access techniques.
4. To understand the functioning of network and transport layer protocols, including addressing, routing, and congestion control.
5. To study application layer protocols, network security mechanisms, and techniques to ensure data integrity.

Course Outcomes:

After completing the course, the student will be able to,

1. Understand of the fundamentals of data communications and computer networks.
2. Learn different network architectures, models, and transmission media used in data communication.
3. Analyze error detection and correction methods, data link protocols, and medium access techniques.
4. Grasp the functioning of network and transport layer protocols, including addressing, routing, and congestion control.
5. Gain knowledge on application layer protocols, network security mechanisms, and techniques to ensure data integrity.

UNIT I

Overview of Data Communication and Networking: Introduction; Data communications: components, direction of data flow; network criteria, physical structure, categories of network (LAN, MAN, WAN); Internet: brief history, Protocols and standards; Reference models: OSI reference model, TCP/IP reference model, their comparative study.

UNIT II

Physical Layer: Overview of data (analog & digital), signal (analog& digital), transmission (analog & digital) & transmission media (guided & unguided), queuing theory , its applications in data communication, Data Encoding Techniques, Circuit switching, time division & space division switching.

UNIT III

Data link Layer: Types of errors, framing (character and bit stuffing), error detection & correction methods; Flow control; Protocols: Stop & wait ARQ, Go-Back- N ARQ, Selective repeat ARQ, HDLC

Medium Access sub layer: Point to Point Protocol, Token Ring; Reservation, Polling, Multiple access protocols: Pure ALOHA, Slotted ALOHA, CSMA, CSMA/CD, CSMA/CA Traditional Ethernet, fast Ethernet.

UNIT IV

Network layer: Internetworking & devices: Repeaters, Hubs, Bridges, Switches, Router, Gateway; Addressing: IP addressing, subnetting; Routing: techniques, static vs. dynamic routing, Unicast Routing Protocols: RIP, OSPF, BGP; Other Protocols: ARP, IP, ICMP, IPV6

Transport layer: Process to Process delivery; UDP; TCP; Congestion Control: Open Loop, Closed Loop choke packets; Quality of service: techniques to improve QoS: Leaky bucket algorithm, Token bucket algorithm.

UNIT V

Application Layer: Introduction to DNS, SMTP, SNMP, FTP, HTTP & WWW; Security: Cryptography (Public, Private Key based), Digital Signature, Firewalls.

Textbooks:

1. B. A. Forouzan – "Data Communications and Networking (3rd Ed.) " – TMH
2. A.S. Tanenbaum – "Computer Networks (4th Ed.)" – Pearson Education/PHI

Reference Books:

1. W. Stallings – "Data and Computer Communications (5th Ed.)" – PHI/ Pearson Education
2. Kurose and Rose – "Computer Networking -A top down approach featuring the internet" – Pearson Education
3. Leon, Garica, Widjaja – "Communication Networks" – TMH